

Part1. Workspace Description

Workspace Title

Clinical characteristics of antidepressant treatment escalation in major depressive disorder in All of Us

Research Use Statement

1) What is the primary purpose of your project?

Research purpose:

Disease-focused educational research

Name of disease:

Major depressive disorder(MDD); treatment-resistant depression(TRD) (proxy based on treatment escalation)

2) Please provide a summary of your research purpose by responding to the questions below.

2.1. What are the specific scientific question(s) you intend to study, and why is the question important?

Major depressive disorder (MDD) is the most common mental disorder, affecting over 300 million people in the world. Around 21 million Americans experience at least one episode of MDD in 2021; one-third suffer from treatment-resistant depression (TRD), defined as a failure to get a response or remission after two or more trials of MDD monotherapy with adequate dose and duration. While the burden of TRD on the U.S. economy was calculated as 43.8 billion in 2021, the reasons for finding a functional combination therapy go beyond the costs, as half of TRD patients report lifetime suicidal ideation. Some people need to switch between multiple antidepressants or require additional medications, such as antipsychotics, lithium or Ketamine/Esketamine, to achieve symptom improvement. Numerous studies have been conducted to find effective treatment strategies in recent years; however, limited treatment strategies are still available due to the complex mechanism of TRD.

Since direct measures of treatment response are not consistently available in electronic health records, this study will use real-world treatment patterns as proxies for poor antidepressant response. This study will examine how participants with antidepressant treatment escalation differ from participants who remain on a single SSRI for an extended period.

The specific questions we will study are:

1. Do patients with antidepressant treatment escalation differ from patients maintained on a single SSRI in demographic characteristics such as age, sex, race and ethnicity, and rural vs urban residence?
2. Do the groups differ in clinical characteristics such as psychiatric comorbidities and medical comorbidities?

3. Do the groups differ in health care utilization, such as the number of psychiatric visits per year, hospitalizations related to MDD, and prescription counts for antidepressants, and do these patterns relate to symptom severity when PHQ-9 data are available?

These findings will provide a clearer picture of real world treatment patterns on depression specially treatment-resistant depression and will support future large scale genome-wide association studies after the phenotyping approach is established.

2.2. What are the scientific approaches you plan to use for your study?

We will use electronic health record and survey data from the All of Us Research Program Controlled Tier. Participants with major depressive disorder (MDD) based on ICD codes (e.g., F32, F33) and EHR diagnosis will be identified using ICD-9 and ICD-10 diagnosis codes, and antidepressant treatment patterns will be derived from prescription records.

We will define two cohorts:

1. Treatment escalation group: patients with ≥ 2 switches between different antidepressant medications or evidence of augmentation therapy such as antipsychotic medications, lithium or ketamine/esketamine.
2. Single treatment group: patients prescribed one SSRI for at least 6 months without switching

Descriptive analyses and visualizations will summarize treatment patterns and individuals characteristics. Logistic regression models will examine associations between treatment escalation and demographic and clinical factors, including age, sex, race and ethnicity, psychiatric comorbid conditions, and rural vs urban residence, and health care utilization measures such as the number and types of visits. Analyses will be conducted using BigQuery and Python/R within the All of Us Researcher Workbench. Limitations include reliance on electronic health record data, which may not fully capture symptom severity, medication adherence, or reasons for treatment changes.

2.3. What are the anticipated findings from the study? How would your findings contribute to scientific knowledge?

We expect that participants with treatment escalation will show higher clinical complexity than those maintained on a single SSRI, including higher rates of psychiatric comorbidities and greater health care utilization. This study will demonstrate how real world prescribing patterns and health care data can be used to characterize depression treatment complexity at scale in All of Us. The results will provide a practical foundation for future genome wide analyses of treatment resistance once cohort definitions and data extraction are validated.

The All of Us Research Program encourages researchers to disseminate their research findings to both scientific and lay/community audiences, to maximize the value of the resource and to recognize the contributions of participant partners. Please tell us how

you plan to disseminate your research findings. Choose as many options below as applicable, and specify details in the text box, if available.

Results from this study will be shared through academic coursework presentations and written reports. Findings may also be summarized in clear, non-technical language suitable for a general audience to help All of Us participants understand how their data contribute to research on depression and mental health treatment.

- 3) The *All of Us* Research Program would like to understand how (or if) your research outcome may fit into the *All of Us* Research Program Scientific Framework. Please select all of the statements below that describe the outcomes you anticipate from your research.**

None of these statements apply to this research project.

- 4) Population of interest**

Will your study focus on any historically underrepresented populations?

Yes, this study will include groups that are historically underrepresented in biomedical research.

If "Yes": please indicate your underrepresented population(s) of interest:

These groups are included in our cohort:

- Older adults aged 65 to 74
- Older adults aged 75 and older
- Participants living in rural or non-metropolitan areas

- 5) Confirm if your study includes research involving American Indian and Alaska Native populations. Refer to the Policy on Respectful Research Involving American Indian and Alaska Native Populations.**

This study is not designed to focus on American Indian or Alaska Native populations. If we include them in our analysis, results will be reported in accordance with All of Us policies to ensure respectful and appropriate interpretation.

- 6) Request for Review of Research Purpose Description**

Would you like to request a review of your research purpose statement by the Resource Access Board? Note: Your response to this question is private and will not be displayed on the Research Hub.

No, I have no concerns at this time about potential stigmatization based on my study.

Part 2. Provide a list of variables they are considering for inclusion in their cohort demographic table. This list must include:

1. Demographic variables:

- Age at index date (first antidepressant prescription)
- Sex at birth
- Race
- Ethnicity
- Rural vs urban residence

2. Clinical variables (diagnosed based on ICD-9/ICD-10):

- Major depressive disorder (inclusion criterion)
- Anxiety disorder or PTSD (clinical covariate)
- Opioid Use Disorder (clinical covariate)
- At least one medical comorbidity such as diabetes or hypertension (clinical covariate)

3. Self-report questionnaire variables:

- PHQ-9 total score (depression severity), if available
- Self-reported family history of depression or psychiatric disorders